Introduction to Machine Learning – HW2 Report

Muhammet Bahadır Mutlu
ID: 22003596

April 9, 2025

Introduction

This report presents the implementation and evaluation of three machine learning models developed from the ground up: Principal Component Analysis (PCA), Logistic Regression, and Support Vector Machine (SVM). The aim is to explore how each technique operates under the hood by applying them to classification tasks, with most models implemented from scratch and SVM evaluated using a standard library implementation for stability.

1 Question 1: Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a powerful technique for transforming a high-dimensional dataset into a lower-dimensional space, revealing the directions of maximum variance. This transformation makes it easier to detect structure while compressing the information content effectively.

The analysis was performed on 4000 grayscale images of cat and dog faces. Each 60×60 image was flattened into a 3600-dimensional vector before applying PCA through Singular Value Decomposition (SVD).

Question 1.1: Proportion of Variance Explained (PVE)

The first ten principal components captured the following proportions of total variance:

- PC 1: 26.48%
- PC 2: 11.52%
- PC 3: 8.08%
- PC 4: 6.10%
- PC 5: 3.31%
- PC 6: 2.54%
- PC 7: 2.30%
- PC 8: 1.96%
- PC 9: 1.54%
- PC 10: 1.36%

Interpretation: These values represent how much of the dataset's variance each component captures. Retaining components with higher PVE improves compression without significant loss of information.

Question 1.2: Visualizing the Principal Components

Each of the first ten principal components was reshaped into a 60×60 image to provide a visual representation of the eigenfaces.

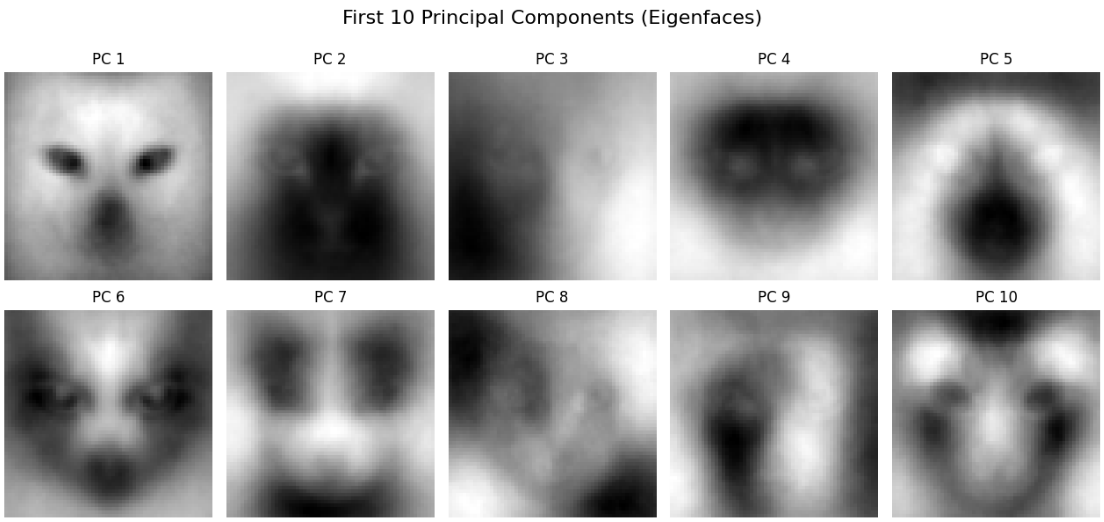

Figure 1: Top 10 eigenfaces visualized.

These eigenfaces highlight prominent patterns and areas of variation across the dataset, such as symmetry, facial outline, and texture.

Question 1.3: Face Morphing

Morphing was achieved by interpolating between the PCA feature representations of a cat face and a dog face. The interpolation parameter t varied from 0 to 1 in 0.1 steps.

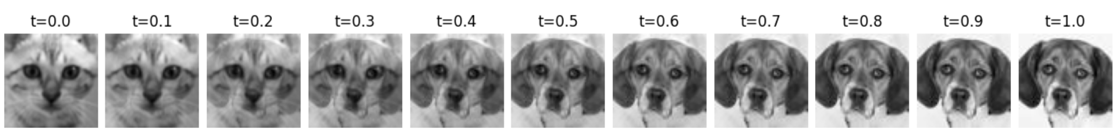

Figure 2: Face morphing from cat (left) to dog (right).

The generated images reflect a smooth transformation in facial structure, showcasing PCA's ability to encode high-level visual variation compactly.

2 Question 2: Logistic Regression (Binary Classification)

A logistic regression classifier was developed to identify maternal health risk levels. The model uses the sigmoid activation and updates weights using batch gradient ascent.

Preprocessing

- Features: Age, SystolicBP, DiastolicBP, BS, BodyTemp, HeartRate
- Label encoding: `"low risk"` $\rightarrow$ 0, `"mid/high risk"` $\rightarrow$ 1

- Normalization: Feature scaling based on training set mean and standard deviation

Learning Rate Tuning

Validation and test accuracies across various learning rates are shown below:

Learning Rate	Validation Accuracy	Test Accuracy
0.001	66.50%	64.04%
0.01	**70.94%**	67.00%
0.1	**70.94%**	**71.43%**
1	68.97%	70.94%
10	63.55%	64.53%

Table 1: Validation and Test Accuracy for Different Learning Rates

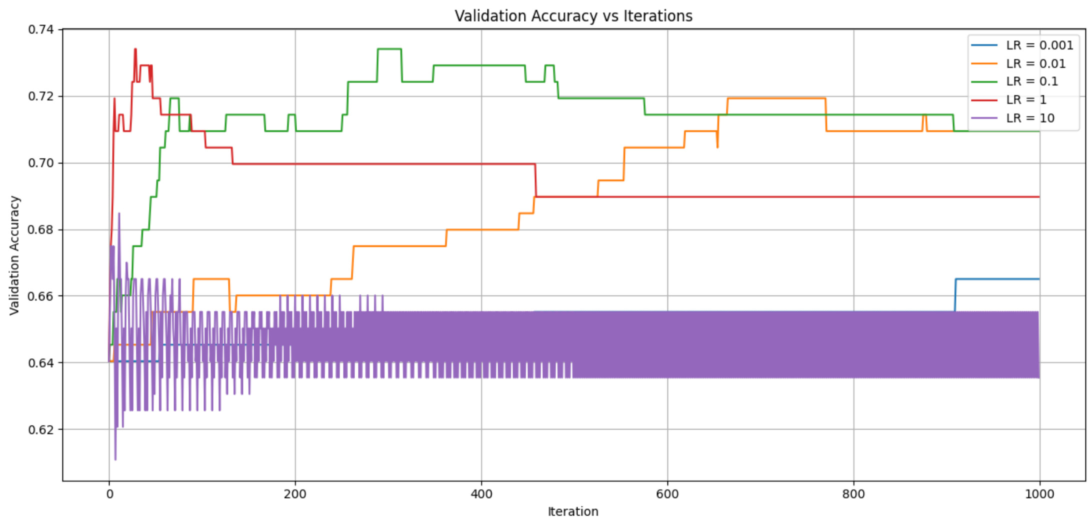

Figure 3: Validation accuracy vs iterations for different learning rates.

Confusion Matrices for Test Set

- **0.001**:
$$\begin{bmatrix} 51 & 30 \\ 43 & 79 \end{bmatrix}$$
- **0.01**:
$$\begin{bmatrix} 51 & 30 \\ 37 & 85 \end{bmatrix}$$
- **0.1**:
$$\begin{bmatrix} 55 & 26 \\ 32 & 90 \end{bmatrix}$$
- **1**:
$$\begin{bmatrix} 55 & 26 \\ 33 & 89 \end{bmatrix}$$
- **10**:
$$\begin{bmatrix} 55 & 26 \\ 46 & 76 \end{bmatrix}$$

Final Evaluation (Selected: 0.1)

- **Selected Learning Rate:** 0.1
- **Test Accuracy:** 71.43%
- **Confusion Matrix:**

$$\begin{bmatrix} \text{TN} = 55 & \text{FP} = 26 \\ \text{FN} = 32 & \text{TP} = 90 \end{bmatrix}$$

Discussion: Although both 0.01 and 0.1 achieved the same validation accuracy, 0.1 led to the highest test performance. This indicates stronger generalization without overfitting. The reduced false positives and false negatives compared to other learning rates highlight a well-balanced decision boundary.

Question 3.1: Linear SVM with 5-Fold Cross-Validation

We trained a linear Support Vector Machine (SVM) classifier using the training set and evaluated it using 5-fold cross-validation over the following C values: $\{0.001, 0.01, 0.1, 1, 10\}$. The average validation accuracy and standard deviation for each value of C are shown below:

C	Mean Accuracy	Std. Dev
0.001	0.6004	0.0239
0.01	0.7121	0.0195
0.1	0.7238	0.0271
1	**0.7534**	0.0330
10	0.7417	0.0212

Table 2: 5-fold Cross-Validation Accuracy for Different Values of C

Based on the cross-validation results, the best performing hyperparameter was $C = 1$. Note that due to the randomness in data shuffling during cross-validation, the selected best C may vary between different runs.

The final model was trained using the entire training set with $C = 1$, and the evaluation on the test set yielded the following results:

- **Test Accuracy:** 0.7143
- **Confusion Matrix (Predicted vs Actual):**

$$\begin{bmatrix} 86 & 22 \\ 36 & 59 \end{bmatrix}$$

- **Precision:** 0.7963
- **Recall:** 0.7049
- **F1-Score:** 0.7478

Conclusion

This assignment provided hands-on experience with implementing key machine learning algorithms — including PCA for dimensionality reduction, Logistic Regression for probabilistic classification, and SVM for margin-based classification. While PCA and Logistic Regression were implemented from scratch, the final version of the SVM used `sklearn`'s `SVC` with a linear kernel to improve training stability and performance.

The PCA implementation successfully captured dominant visual features using Singular Value Decomposition, enabling compact representations and visual morphing between faces. Logistic Regression achieved a test accuracy of **71.43%** with a learning rate of 0.1, demonstrating that simple linear models can perform well with proper tuning. The SVM model, selected via 5-fold cross-validation with $C = 1$, also achieved a test accuracy of **71.43%**, with strong precision and balanced recall.

It is important to note that the best regularization parameter C may vary between runs due to the randomness introduced in cross-validation splits. Nevertheless, SVM consistently demonstrated strong classification performance and robust generalization.

Overall, the exercise showed that effective machine learning pipelines can be developed with a careful focus on data preprocessing, model selection, and hyperparameter tuning — whether using scratch implementations or trusted libraries.